

Pipeline Methodology and Results Report

2026-08-01

1 Scope

This report describes how the current LLM-assisted entity resolution (ER) pipeline works end to end, explains each attribute-selection methodology it implements, and compares results across those methodologies on three benchmark datasets: Fodors–Zagat, DBLP–ACM, and Amazon–Walmart.

All numbers reported here come from run logs that match the pipeline’s current configuration: SentenceBERT embedding model `BAAI/bge-large-en-v1.5`, matcher LLM `google/gemini-2.5-flash` (via OpenRouter), and the per-dataset blocking and adaptive hyperparameters currently defined in `code/main.py`. A number of older logs were excluded because they do not reflect the current pipeline: runs embedded with a superseded model (`all-mpnet-base-v2`), runs matched with a different LLM (`deepseek/deepseek-v4-flash` or `google/gemini-2.0-flash-001`), pre-refactor logs that predate the current run-summary schema, one truncated Fodors–Zagat run that only produced 1,105 of 2,665 expected result pairs, and two stale aggregate CSVs (`logs/experiment_summary.csv` and `logs/step1_metrics_data.csv`) that were both generated under different blocking hyperparameters than the pipeline currently uses. Excluding these keeps every number in this report directly comparable.

One further exclusion is specific to Fodors–Zagat. During analysis its `class` attribute was found to be the ground-truth cluster identifier (Section 5.5), which makes any result computed with that attribute in scope uninterpretable. All Fodors–Zagat numbers in this report therefore come from a second generation of runs executed with `--exclude-attributes class`; the earlier Fodors–Zagat runs are retained only in Section 5.5, where they are used to quantify the effect of the leak. DBLP–ACM was checked and is clean. Amazon–Walmart’s `original_id` attribute was also checked and does *not* leak (it disagrees on 100% of ground-truth pairs), though it is uninformative by construction and is a candidate for exclusion in future runs.

A third exclusion applies to DBLP–ACM. Of the thirteen runs in the reseeding sweep attempted on 2 August, eight failed on every LLM call and a ninth failed on 28.9% of them, the API budget having been exhausted mid-sweep. Because the pipeline scores a failed call as a non-match and still writes complete metrics, those nine runs report numerically valid but meaningless results; all nine are excluded, leaving the five `random` runs analysed in Section 5.1. A later Fodors–Zagat Step-1 run failed the same way and is likewise excluded.

Finally, a note on the report’s status. Two defects in the measurement apparatus were identified during this analysis and have since been corrected in the pipeline: the prompt field order was uncontrolled (Section 5.2), and the LLM client declared no token cap (Section 5.3). Every result reported here predates both corrections. The report is therefore a complete account of what the current evidence supports, together with an explicit statement of which conclusions those corrections put back in question; Section 9 sets out what remains to be run.

2 Pipeline Overview

The pipeline runs in five stages, each timed and recorded in a per-run JSON summary under `logs/<dataset>/runs/`.

Load and normalize. Both tables and the ground truth are read, ID columns are dropped, and every remaining field is passed through a normalization step that expands or standardizes address and directional abbreviations, and cleans up Unicode/HTML artifacts, so that superficial formatting differences do not by themselves create false mismatches later in the pipeline.

Embed and block. Each attribute column is encoded independently with the SentenceBERT model, the column vectors are concatenated and L2-normalized into one embedding per record, and multi-probe locality-sensitive hashing (random hyperplanes, with single-bit flips at query time) retrieves the top- k most similar Table-B candidates for every Table-A record. This reduces the pairwise comparison space from the full cross product down to a much smaller candidate-pair set before any LLM is involved.

Step 1 – global tuple condensation (optional). One attribute subset is chosen once for the whole dataset and applied to every candidate pair before re-blocking on the condensed attributes. This step is described in full in Section 3.1.

Step 1.5 – adaptive per-pair attribute selection (optional). Instead of one global subset, a small sample of candidate pairs is profiled by the LLM for per-attribute importance, and a hybrid predictor/retriever model transfers a *different* attribute subset to every candidate pair. This step is described in full in Section 3.2.

Match. One LLM call decides Yes/No per candidate pair, using only the attributes selected for that pair (all attributes if neither Step 1 nor Step 1.5 ran). Every response is cached on disk keyed by a hash of the model, prompt template, and record content, so repeated experiments over the same configuration do not re-spend tokens.

Metrics and logging. Precision, recall, and F1 are computed both at the candidate-pair level (only pairs that survived blocking) and end-to-end (relative to all ground-truth matches, so blocking misses count against recall), alongside token-usage accounting split by stage.

3 Methodology

3.1 Step 1: Global Tuple Condensation

The motivation for Step 1 is that showing an LLM every attribute for every pair wastes tokens on fields that are frequently empty, redundant, or simply irrelevant to the match decision, and can make the model’s decision worse by diluting the signal from the few attributes that actually distinguish entities. Step 1 picks *one* attribute subset for the whole dataset, used for every pair, via one of four strategies.

Manual selection. A fixed, user-supplied list of attribute names is kept and everything else is dropped, with no data-driven reasoning involved. It is the cheapest strategy to run (no LLM calls, no labeled data required) and is appropriate when a person can already tell from the schema which fields carry identity information, but it does not adapt if that judgment is wrong for some records.

Heuristic selection. Attributes are pruned using two label-free, corpus-level statistics: the fraction of missing values (an attribute that is empty in most records cannot be evidence of anything) and the fraction of unique values (an attribute that takes almost the same value everywhere carries little discriminative signal). Attributes that survive pruning are also *compressed*: long text fields

are stripped of boilerplate words (via a stopword/marketing-language regex list) and only high-signal tokens are kept – numbers, alphanumeric codes/identifiers, capitalized proper nouns, and unit-bearing measurements (e.g. "12 kg") – so that a lengthy description field still contributes its distinctive tokens without its full token count.

LLM-guided selection. A small labeled sample of matching and non-matching pairs is shown to the LLM one pair at a time, together with the question “which fields were most influential in deciding whether these two records match?”, and the model returns the smallest sufficient subset of attribute names for that pair. Responses are aggregated across the sample by *selection frequency* per attribute (what fraction of responses named that attribute), attributes are ranked by that frequency, and the final subset keeps every attribute above a frequency threshold (optionally capped to a top- k).

Supervised selection. For the same labeled sample, three field-level similarity features are computed per attribute for every pair – token-set Jaccard overlap, normalized edit-distance similarity, and an exact-match indicator – and a logistic regression classifier (features standardized, class-balanced) is trained to predict match/non-match from all attributes’ features at once. Each attribute’s importance is the maximum absolute regression weight across its three features, normalized so all attributes’ importance scores sum to one; the top-ranked attributes above a threshold (optionally capped to top- k) are kept. Because it learns from labeled evidence directly, this strategy tends to identify a very small, highly discriminative attribute subset (see Section 5).

Whichever strategy is used, the condensed table is then re-embedded and re-blocked (so blocking itself also benefits from the smaller, more focused attribute set) before matching.

3.2 Step 1.5: Pair-Adaptive Attribute Selection

Step 1.5 addresses a limitation of Step 1: even a well-chosen global attribute subset is not equally informative for every pair. For one pair an exact model number may already be decisive; for another, the deciding signal might be an author mismatch or a subtle title variation, while the model number field is missing or uninformative. Step 1.5 replaces one global subset with a *per-pair* subset, produced in two stages.

Profiling stage. A small sample of post-blocking candidate pairs (the “profile pairs”) is selected *without using ground-truth labels* by one of five samplers (described below). For each profile pair, the LLM is asked to score every attribute’s usefulness for the match/non-match decision on an integer 0–3 scale (0 = not useful/missing/generic, 3 = highly important or decisive), and the raw scores are normalized per pair into an importance vector that sums to one.

Transfer stage. A `HybridAttributeSelector` is fit on the profiled pairs’ label-free similarity features (token overlap, edit similarity, exact match, semantic similarity, numeric similarity, and an is-numeric indicator, computed per attribute) against their LLM-scored importance vectors. It combines two sub-models: a Ridge-regression *predictor* that maps a pair’s features directly to a predicted importance vector, and a cosine-similarity *retriever* that finds the k most similar profiled pairs and averages their importance vectors. The two signals are combined by a weighted average (0.5/0.5 in the current configuration). For every candidate pair (profiled or not), attributes are then selected in descending combined-importance order until either a fixed top- k is reached or cumulative importance crosses a threshold (0.8 in the current configuration, bounded between a minimum and maximum attribute count), with a fallback rule that force-includes at least one attribute tagged with a required “identity” role (e.g. title, name, model number) even if the learned ranking would otherwise drop it.

The five samplers below differ only in *which* candidate pairs get profiled by the LLM; the transfer stage is identical for all of them.

Random sampling. Profile pairs are drawn uniformly at random from all candidate pairs. This is the label-free baseline against which the other samplers are compared.

Similarity-stratified sampling. Candidate pairs are bucketed into low/medium/high terciles by their blocking-stage cosine similarity, and profile pairs are drawn from each bucket in fixed proportions (60% medium, 20% low, 20% high in the current configuration). This intentionally profiles some clearly similar pairs, some clearly dissimilar pairs, and a majority of ambiguous near-threshold pairs, on the reasoning that ambiguous pairs are where attribute importance is hardest to guess and most useful to learn from.

Diversity sampling. Profile pairs are chosen by farthest-first traversal in the label-free pair-feature space: starting from the pair farthest from the sample centroid, each subsequent pair is the one that maximizes its minimum distance to all previously chosen pairs. This favors broad coverage of distinct similarity “shapes” (e.g. pairs that agree on names but disagree on numbers, versus pairs that disagree on everything) over any particular similarity level.

Stratified diversity sampling. The same low/medium/high similarity strata as similarity-stratified sampling are used, but within each stratum, pairs are chosen by the same farthest-first diversity rule rather than uniformly at random. This is meant to combine both benefits: similarity-level balance and within-level feature diversity.

Importance-based sampling. Pairs are ranked by a label-free *importance proxy score* that estimates, without ground truth, how informative a pair is likely to be for learning attribute importance:

$$\text{proxy} = 0.50 \cdot \text{evidence_contrast} + 0.30 \cdot \text{similarity_uncertainty} + 0.20 \cdot \text{evidence_mixedness}$$

where `evidence_contrast` is the standard deviation of per-attribute agreement signals (favoring pairs where attributes disagree with each other, rather than all pointing the same way), `similarity_uncertainty` favors pairs whose blocking similarity is far from the sample’s typical value, and `evidence_mixedness` favors pairs whose average per-attribute agreement sits near 0.5 (neither clearly matching nor clearly non-matching). The top-scoring pairs (with random tie-breaking) become the profile sample.

4 Experimental Setup

Table 1 lists the current per-dataset configuration used for every number in this report. Table 2 lists the exact run directory each result was read from, for traceability.

5 Results

Table 3 compares the five Step-1.5 profile samplers on DBLP–ACM. Blocking pair completeness is 1.0000 for every row shown (every ground-truth match survived blocking), so candidate-pair metrics and end-to-end metrics are identical here. Bold marks the best value per column.

Dataset	Blocking (tables/planes/flips/top- k)	Profile size	Attrs. available
Fodors-Zagat	5 / 8 / 1 / 5	20	5 (6 less class)
DBLP-ACM	15 / 8 / 1 / 5	50	4
Amazon-Walmart	10 / 4 / 1 / 4	50	13

Table 1: Current per-dataset blocking and Step-1.5 profiling configuration. Adaptive selection policy is shared across datasets: top- k retrieval 5, min/max attributes 2/5, cumulative-importance threshold 0.8, required role identity.

Dataset	Method	Run directory
DBLP-ACM	step15 / random	20260615_120346_step15_random
DBLP-ACM	step15 / similarity_stratified	20260615_121510_step15_similarity_stratified
DBLP-ACM	step15 / diversity	20260615_123251_step15_diversity
DBLP-ACM	step15 / stratified_diversity	20260615_133158_step15_stratified_diversity
DBLP-ACM	step15 / importance_based	20260615_221942_step15_importance_based
DBLP-ACM	step1 / supervised	20260615_225756_step1
DBLP-ACM	step15 / random, 5 seeds	20260802_035813-20260802_044306 (5 dirs)
DBLP-ACM	<i>aborted, API exhaustion</i>	20260802_045249-20260802_051239 (8 dirs)
Fodors-Zagat	step15 / all 5 samplers, 6 runs each	20260801_2246*-20260802_0032* (30 dirs)
Fodors-Zagat	aggregated in	logs/Fodors-Zagat/sampler_seed_sweep.json
Fodors-Zagat	step1 / supervised	20260801_230544_step1
Fodors-Zagat	full / no selection	20260801_230923_full
Fodors-Zagat	full / no selection, replicate	20260802_173229_full
Amazon-Walmart	step15 / similarity_stratified	20260615_163220_step15_similarity_stratified

Table 2: Provenance: every metric in this report is read from the `run_summary.json` in the listed `logs/<dataset>/runs/` directory. All Fodors-Zagat runs listed were executed with `--exclude-attributes class`; all runs listed were executed with `--disable-matcher-cache`. The eight aborted DBLP-ACM directories are recorded for completeness but contribute no metrics; see Section 5.1.

Dataset	Sampler	Attrs	F1	Precision	Recall	TP	FP	FN
DBLP-ACM	random	4	0.9235	0.9722	0.8795	1,956	56	268
DBLP-ACM	similarity_stratified	4	0.9200	0.9715	0.8737	1,943	57	281
DBLP-ACM	diversity	4	0.9162	0.9761	0.8633	1,920	47	304
DBLP-ACM	stratified_diversity	4	0.9066	0.9669	0.8534	1,898	65	326
DBLP-ACM	importance_based	4	0.9605	0.9768	0.9447	2,101	50	123

Table 3: Step-1.5 profile-sampler comparison on DBLP-ACM, current pipeline (BAAI/bge-large-en-v1.5 + google/gemini-2.5-flash). Single run per sampler at seed 42. Section 5.1 shows that this single-seed ranking is not reliable.

5.1 Reseeding DBLP-ACM: A Partial Sweep

A repeated-seed sweep was attempted on DBLP-ACM to test whether the ranking in Table 3 survives reseeding. It completed only in part. The five `random` runs (seeds 42–46) finished cleanly with zero API errors. The sixth run, `similarity_stratified` at seed 42, failed on 3,785 of its 13,080 pairs (28.9%), and the remaining eight runs failed on all 13,080 pairs, consistent with the OpenRouter budget being exhausted mid-sweep. The pipeline scores a failed LLM call as a non-match and still writes a complete `run_summary.json`, so those nine runs carry plausible-looking but meaningless metrics (the fully-failed runs report $F1 = 0.0000$; the partial run reports 0.7672). They are excluded throughout. Only the `random` rows in Table 4 are evidence.

Seed	F1	Precision	Recall	TP	FN
42	0.9258	0.9718	0.8840	1,966	258
43	0.9170	0.9766	0.8642	1,922	302
44	0.8847	0.9710	0.8125	1,807	417
45	0.9150	0.9760	0.8611	1,915	309
46	0.9119	0.9749	0.8566	1,905	319
mean	0.9109	0.9741	0.8557	1,903	321
SD	0.0155	0.0025	0.0264	59	59
range	0.0411	0.0056	0.0715	159	159

Table 4: DBLP-ACM, `random` profile sampler over five seeds. Blocking is held fixed across all five runs, so the candidate-pair set is identical and the only varying input is which 50 pairs are profiled.

The spread is large. Reseeding the profile sample alone moves end-to-end F1 by 0.0411 and true positives by 159 — comfortably wider than the 0.0370–0.0539 gap that separates `importance_based` from the other four samplers in Table 3. A single-seed sampler ranking on this dataset is therefore not interpretable.

This is not LLM non-determinism. The June and August `random` runs at seed 42 share identical configuration and produce identical attribute selections on 99.9% of pairs; their answers differ on 0.22% of pairs, worth 10 true positives and 0.0023 F1. Replicate noise is roughly one-eighteenth of the seed-to-seed spread.

5.2 Prompt Attribute Order Is an Uncontrolled Variable

Decomposing the seed-to-seed disagreements locates the cause, and it is not the one the experiment was designed to measure. `HybridAttributeSelector.select_attributes` ranks attributes by predicted importance (`np.argsort(-combined)`) and emits the selected subset *in that order*. `code/matcher.py:116` rebuilds each record as a dict comprehension over that list and serializes it

with `json.dumps`, so the ranking order is reproduced verbatim as the field order in the prompt. Because predicted importance shifts continuously with the profile sample, reseeding permutes the prompt even when it selects exactly the same attributes.

Table 5 conditions every cross-seed pair comparison on whether the two runs selected the same attribute *set* and the same *order*.

Condition	Mean pairs per seed comparison	Answers disagree
Same attribute set, same order	1,871	0.10%
Same attribute set, different order	4,772	5.44%
Different attribute set	6,437	1.23%

Table 5: DBLP-ACM, `random` sampler, all ten pairwise comparisons among the five seeds (13,080 candidate pairs each). Holding the selected attribute set fixed and permuting only the order in which those attributes appear in the prompt changes the matcher’s answer 54 times more often than presenting the identical prompt twice.

Presenting the identical prompt twice flips 0.10% of answers; presenting the same attributes in a different order flips 5.44%. Changing *which* attributes are selected flips 1.23% — less than reordering does, though that group is not a controlled comparison, since the pairs on which two selectors disagree are also the pairs where they were least confident. The controlled contrast is the first two rows, and it is unambiguous: on this schema the matcher is more sensitive to field order than to field choice.

Order also varies far more than set does. Across seed comparisons the two runs agree on the attribute set for 42–67% of pairs, but among those agreements the order still differs on 51–96%. DBLP-ACM has only four attributes, and the selector is in practice making a single binary decision per pair — keep `year` or not (mean selection size 3.51–3.79, never below 3 or above 4). The remaining variation between runs is permutation.

The consequence is that Step-1.5’s measured effect on DBLP-ACM is confounded. Each run varies two things at once: the attribute subset, which is the intended independent variable, and the prompt field order, which is an unintended by-product of ranking and which by this measurement is the stronger of the two. The same confound is present in every Step-1.5 number in this report, Fodors-Zagat included, and has not been quantified there.

This has since been corrected in the pipeline. Both selection stages now sort their chosen subset into table-column order before condensation, so field order is fixed across runs and attribute choice is the only thing a reseed varies; the importance ranking is still recorded, but no longer reaches the prompt. Runs produced after the fix carry `canonical_attribute_order` in their run summary, which distinguishes the two generations. Every result in this report predates the fix, so the confound stands as described above; re-running the comparisons is the first item of remaining work (Section 9).

5.3 An Unset Token Cap Made Runs Fail on Budget

A second infrastructure defect surfaced while attempting the re-runs and is recorded here because it explains the aborted sweeps of Section 5.1. The shared OpenRouter client never set `max_tokens`, so each request defaulted to the model’s full output ceiling of 65,535 tokens. OpenRouter reserves credit against that declared ceiling rather than against actual usage, and rejects a request with HTTP 402 whenever the remaining balance cannot cover it.

The reservation was pathologically conservative. Measured over clean runs, the matcher emits *exactly one* completion token on every call — 13,080 of 13,080 on DBLP-ACM, 2,665 of 2,665 on Fodors-Zagat — because it answers with a single label. Attribute-importance profiling emits 50 tokens on DBLP-ACM’s four-attribute schema and 133 on Amazon-Walmart’s thirteen. Every call therefore reserved roughly four orders of magnitude more budget than it consumed, and runs began failing while the account still held usable credit.

The client now sends an explicit cap: 8 tokens for matching, 512 for profiling. A post-fix `--mode full` run on Fodors-Zagat completed with zero API errors and 2,665 completion tokens over 2,665 calls, confirming the cap is not truncating responses. This changes no experimental result — the caps sit far above observed usage — but it removes a failure mode that silently corrupted nine runs.

Fodors-Zagat was run six times per sampler (seeds 42–46, plus one repeat of seed 42), so it is reported as a distribution rather than a point estimate in Table 6. This matters: on a 110-pair ground truth a single true positive is worth 0.9 percentage points of recall, and the single-seed figures that this report previously carried turned out not to be representative for two of the five samplers.

Sampler	Mean F1	SD	Min	Max	TP range	Mean prec.	Mean rec.
diversity	0.9222	0.0021	0.9179	0.9231	95–96	0.9796	0.8712
stratified_diversity	0.9152	0.0042	0.9135	0.9238	95–97	0.9695	0.8667
similarity_stratified	0.9110	0.0211	0.8788	0.9384	87–99	0.9827	0.8500
random	0.8843	0.0429	0.8316	0.9275	79–96	0.9854	0.8045
importance_based	0.8356	0.0063	0.8254	0.8437	78–81	0.9876	0.7242

Table 6: Fodors-Zagat (`class` excluded), Step-1.5 profile samplers over six runs each. End-to-end metrics; blocking pair completeness is 1.0000 throughout, and blocking is held fixed across seeds so that only the profiling sample varies.

Amazon-Walmart only has one Step-1.5 run on the current pipeline (`similarity_stratified`), so it is reported separately rather than alongside the five-way comparison above. Pair completeness is 0.9099, so 103 ground-truth matches were already lost during blocking, before the LLM ever saw them; candidate-level and end-to-end metrics therefore differ.

Metric level	Attrs	Precision	Recall	F1	TP	FN
Candidate-pair	13	0.8588	0.8596	0.8592	894	146
End-to-end	13	0.8588	0.7822	0.8187	894	249

Table 7: Amazon-Walmart, Step-1.5 `similarity_stratified` sampler (only current-pipeline run available). End-to-end FN includes the 103 ground-truth matches lost at blocking.

Table 8 compares Step 1’s supervised global condensation against the best Step-1.5 sampler on each of the two datasets where both a current Step 1 and a current Step 1.5 sweep exist.

Fodors-Zagat is the only dataset with a no-selection baseline (`--mode full`), which makes it the only place where the absolute value of doing attribute selection at all can be measured. Table 9 places that baseline alongside the two selection mechanisms.

The no-selection baseline has since been run a second time under an identical configuration, giving an independent estimate of the matcher’s replicate noise on this dataset. The two runs return F1 of 0.7849 and 0.7826 (73 and 72 true positives, 3 and 2 false positives) on identical token spend. Because `--mode full` applies no selection, the canonical-order fix of Section 5.2 cannot

Dataset	Method	Attrs used	F1	Precision	Recall	Tokens
DBLP-ACM	Step 1, supervised (authors, title)	2	0.9515	0.9289	0.9753	3,502,759
DBLP-ACM	Step 1.5, importance-based (best)	4	0.9605	0.9768	0.9447	3,910,646
Fodors-Zagat	Step 1, supervised (phone, name)	2	0.9537	0.9717	0.9364	658,185
Fodors-Zagat	Step 1.5, diversity (best mean of 6 runs)	5	0.9222	0.9796	0.8712	775,828

Table 8: Global (Step 1) versus adaptive per-pair (Step 1.5, best sampler) attribute selection. F1/precision/recall are end-to-end.

Method	Attrs/pair	F1	Precision	Recall	Tokens	vs. no selection
No selection (full)	5.00	0.7849	0.9605	0.6636	802,477	—
Step 1.5, diversity (mean of 6)	4.00	0.9222	0.9796	0.8712	775,828	+0.1373 F1, −3.3% tokens
Step 1, supervised	2.00	0.9537	0.9717	0.9364	658,185	+0.1688 F1, −18.0% tokens

Table 9: Fodors-Zagat (**class** excluded): attribute selection versus matching on all attributes. F1/precision/recall are end-to-end. “Attrs/pair” is the mean number of attributes shown to the matcher LLM per candidate pair.

affect it, and the 0.0023 gap is attributable entirely to LLM non-determinism. One true positive out of 110 is consistent with the 0.10–0.22% per-pair disagreement measured on DBLP-ACM, and it sets the scale against which the Fodors-Zagat sampler differences in Table 6 should be read.

5.4 Step 1 and Step 1.5 Do Not Operate Under the Same Supervision

Every head-to-head between the two mechanisms in this report compares a method that consumes ground-truth labels against one that does not, and the gap in supervision is wide enough to change what the comparison establishes.

Step 1’s supervised strategy calls `build_step1_labeled_pairs` (`code/main.py:144`), which draws its positive examples directly from the gold set and its negatives from blocked non-matches, then trains a logistic regression on the result. The LLM-guided strategy consumes the same labeled sample. Step 1.5 uses no labels anywhere: profile pairs are chosen by label-free samplers over label-free pair features, and are scored by the LLM rather than against ground truth.

Mechanism	Gold labels consumed	Fodors-Zagat	DBLP-ACM
Step 1, supervised / LLM-guided	50 pos. + 50 neg.	45.5% of gold	2.2% of gold
Step 1, manual / heuristic	none	—	—
Step 1.5, all five samplers	none	—	—

Table 10: Ground-truth consumption by attribute-selection mechanism. Percentages are the 50 sampled positives as a share of each dataset’s gold matches (110 and 2,224 respectively).

On Fodors-Zagat the asymmetry is severe. The ground truth contains 110 matches and the supervised strategy is fitted on 50 of them — it sees 45.5% of the answer key before selecting an attribute. On DBLP-ACM the same 50 positives are only 2.2% of a much larger gold set, and the concern is correspondingly milder.

There is a second issue beyond the asymmetry. The labeled sample is not held out: `build_step1_labeled_pairs` draws from `gt_set`, and the evaluation at `code/main.py:1032` scores against that same `gt_set` without excluding the sampled pairs. Part of the Step-1 score is therefore measured on pairs whose attribute selection was fitted with knowledge of their labels.

This is not catastrophic — what is fitted is a global attribute ranking, not a per-pair decision, so the leak is indirect — but on Fodors–Zagat it touches 45.5% of the evaluation set and should be stated rather than left implicit.

Neither point makes the Step-1 numbers wrong, and neither rescues Step 1.5, which must still demonstrate a win on its own terms. What they change is the question. Step 1.5 is not competing with a peer; it is competing with a method handed a large share of the ground truth, which makes Step 1 supervised closer to an approximate upper bound than to a deployable baseline — in any real deployment the labels it consumes are precisely what the pipeline exists to produce. Read that way, 0.9222 against 0.9537 on Fodors–Zagat under zero supervision is a materially different result from the same gap between two comparable methods.

The genuine like-for-like baselines are Step 1’s *manual* and *heuristic* strategies, both label-free. Neither has a current-pipeline run on any dataset (Section 7), so the evaluation as it stands cannot answer the question the thesis most needs answered.

5.5 Label Leakage in Fodors–Zagat

The Fodors–Zagat schema contains an attribute named `class` which, on inspection, is the ground-truth cluster identifier rather than a data field: `class` takes the same value on both sides for 110 of 110 ground-truth pairs and is unique within each table. Matching on `class` alone therefore solves the dataset exactly, and any method that selects it is being evaluated on a label rather than on evidence.

The Step-1 supervised strategy selected it. Its ranking assigned `phone` and `class` the identical importance score 0.2825, the expected signature of two perfectly predictive features, and the resulting run reported 0.9725 F1 – the highest figure of any method on any dataset in the first generation of runs. That number is not a valid measurement of the method.

The Step-1.5 path was affected differently. The profiling LLM consistently rated `class` the *least* useful attribute of the six (mean importance 1.35 and 1.50 out of 3 across the two runs, against 3.00 for `phone`), so it declined to exploit the leak. The leak nonetheless degraded Step 1.5 by occupying a selection slot with a field the matcher could not interpret: after removing `class`, four of five samplers improved substantially, as shown in Table 11.

Method	F1 with <code>class</code>	F1 without <code>class</code>	Δ
Step 1.5, random	0.7802	0.9179	+0.1377
Step 1.5, similarity_stratified	0.9171	0.9231	+0.0060
Step 1.5, diversity	0.8254	0.9231	+0.0977
Step 1.5, stratified_diversity	0.8108	0.9135	+0.1027
Step 1.5, importance_based	0.8601	0.8377	−0.0224
Step 1, supervised	0.9725 [†]	0.9581	−0.0144

Table 11: Effect of removing the leaking `class` attribute on Fodors–Zagat. F1 is candidate-level for comparability. [†]This figure is contaminated: the supervised strategy selected `class` itself. With `class` removed it selects (`phone`, `name`) instead.

Two consequences follow. First, the first-generation Fodors–Zagat sampler ranking was largely an artifact: removing the leak changes which sampler wins and compresses the spread between them from 0.137 F1 to 0.085. Second, the qualitative conclusion that global condensation outperforms adaptive selection on this dataset survives the correction – Step 1 still leads, but by 0.031 F1 rather than 0.055, and now on a defensible attribute pair.

6 Interpretation

6.1 Fodors–Zagat

The six-run sweep separates two effects that a single run confounds: how accurate a sampler is on average, and how much its accuracy depends on which pairs it happens to draw.

On the first, only one difference is unambiguous. Importance-based sampling is significantly worse than all four other samplers (Welch t from -2.75 against random to -25.63 against stratified diversity), and it is worse by a wide margin: its best run of six (0.8437) falls below the worst run of every sampler except random. The remaining four are statistically indistinguishable from each other in mean, with one marginal exception (diversity over stratified diversity, $t = 3.64$, but a difference of only 0.0070 F1). The apparent ordering among those four in any single run is not a finding.

On the second, the differences are large and systematic, and they track exactly which samplers are stochastic. Repeating a run at a fixed seed and configuration changes the result by 0–3 true positives, which bounds the matcher LLM’s own non-determinism. Against that floor, the deterministic samplers barely move across seeds (diversity SD 0.0021, stratified diversity 0.0042, importance-based 0.0063), whereas the two that genuinely resample vary by an order of magnitude more (similarity-stratified SD 0.0211, random SD 0.0429). A single run of random sampling can land anywhere between 79 and 96 true positives – a span that covers almost the entire distance between the best and worst samplers. Random and similarity-stratified sampling are therefore not merely less accurate on average; they are unreliable when only one run is affordable, which is the operating regime Step 1.5 is designed for.

This reframes what the sampler comparison contributes. The useful property of diversity and stratified-diversity sampling is not a higher mean but a much lower variance: by choosing profile pairs deterministically from the feature geometry rather than by drawing them, they remove the dominant source of run-to-run instability. That the single best individual run in the sweep belongs to similarity-stratified sampling (0.9384) does not contradict this; it is what a high-variance sampler looks like when a draw goes well.

The mean differences that do exist reduce to a single binary decision. Every sampler selects **name**, **addr**, and **phone** for essentially every pair; the only variation is whether the fourth slot goes to **city** or **type**. On the 110 true matches, random, diversity, and stratified-diversity choose **city** for 96–100% of pairs, similarity-stratified for 86%, and importance-based for only 22%. That substitution is what separates the samplers: across the five seed-42 runs, true matches shown (**addr**, **name**, **phone**, **city**) are missed 9.5–12.7% of the time, whereas true matches shown (**addr**, **name**, **phone**, **type**) are missed 29–67% of the time. **type** is a low-cardinality cuisine category whose taxonomy differs between the two sources, so it disagrees on genuine matches and pushes the matcher toward “No”.

Importance-based sampling is the clear loser here (mean 0.8356 F1 over six runs, 29 to 32 false negatives against 13–15 for the others) purely because it makes that substitution most often, and it does so consistently: because its profile selection is deterministic, the poor choice is reproduced on every seed rather than being a bad draw. Notably, this is not a transfer-stage failure in the mechanical sense: leave-one-out diagnostics on its own profile set give it the *highest* internal fidelity of any Fodors–Zagat run (rank correlation 0.82 and selected-set Jaccard 0.96 between predicted and LLM-assigned importance). The transfer stage reproduces its profile sample faithfully; the profile sample is simply unrepresentative, because the proxy score deliberately selects high-contrast, contested pairs, and on those pairs **type** does carry apparent signal.

Against the no-selection baseline the case for doing selection at all is strong. Matching on all

five attributes yields only 0.7849 F1 (recall 0.6636); every selection method beats it by 0.13–0.17 F1, and even the single worst run in the entire six-seed sweep (0.8316) beats it. The token picture is less flattering to Step 1.5: dropping one of five attributes per pair saves only 3.3% of tokens, because the attribute it drops (`city` or `type`) is among the shortest fields. Step 1’s two-attribute global subset saves 18.0% by comparison. Attribute count and token count are not proportional, and Step 1.5’s current selection policy optimizes the former.

6.2 DBLP–ACM

At single seed, DBLP–ACM appeared to be the dataset where Step 1.5 worked best: importance-based sampling reached 0.9605 F1 against 0.90–0.92 for the other four samplers, driven almost entirely by recall (0.9447 versus 0.85–0.88), and edged out Step 1’s supervised condensation (0.9515 F1 on authors + title alone) at the cost of roughly 12% more matching tokens. That reading no longer holds up.

The reseeding sweep (Section 5.1) shows that changing the profile sample alone moves `random`’s F1 across 0.8847–0.9258, a 0.0411 range. The gap the single-seed table attributes to sampler choice is 0.0370–0.0539. The measurement is not sharp enough to resolve the effect it was reporting, so the headline that importance-based sampling wins on DBLP–ACM must be withdrawn pending repeated seeds — not contradicted, but unsupported. It may well survive: it is a deterministic sampler, and the one comparison available at fixed configuration (June versus August `random` at seed 42) puts replicate noise at only 10 true positives. But that has not been shown.

Two structural facts limit what Step 1.5 can express here in any case. First, the schema has four attributes and the policy floor is `min k = 2` with `title` and `authors` selected essentially always, so the selector’s entire per-pair decision reduces to whether to include `year`: observed selection sizes are 3 or 4 and nothing else. Second, `year` agrees on 100% of the 2,224 ground-truth matches, making it a low-information field — including or excluding it should not, and largely does not, change the answer. The apparent recall differences between the $k = 3$ and $k = 4$ groups within a run (0.84–0.98 versus 0.81–0.90) reverse direction between seeds and reflect which pairs the selector routed where, not a causal effect of dropping `year`.

What does move the answers is prompt field order (Section 5.2). On a four-attribute schema where the attribute set is nearly fixed, permutation is most of what reseeding actually varies, and it flips 5.44% of answers. The DBLP–ACM Step-1.5 results measure that confound at least as much as they measure attribute selection.

6.3 Amazon–Walmart

Amazon–Walmart is the largest and most attribute-rich of the three datasets (13 common attributes, 88,296 candidate pairs), and it is also the only dataset where blocking itself is imperfect on the current pipeline (91.0% pair completeness, losing 103 of 1,143 true matches before the LLM is ever consulted). The one current Step-1.5 run available reaches 0.8592 candidate-level F1 but only 0.8187 end-to-end F1, because of that blocking loss. The learned selector keeps all 5 of the allowed maximum attributes for every one of the 88,296 pairs (selection size min/mean/max all equal 5.0). This is not a property of the selector but of the selection policy: normalizing importance across 13 attributes spreads the mass thinly enough that the cumulative-importance target of 0.8 cannot be met within the 5-attribute cap. Using the LLM’s own profiled importance vectors as a best case, the top 5 attributes carry on average 0.612 of the mass and never more than 0.789, and reaching 0.8 would take 7.7 attributes on average. The cap therefore binds unconditionally, and the transfer stage is only choosing *which* five attributes, never how many. Whatever this run

measures, it is not adaptive selection in the sense Step 1.5 intends, and the result should not be read as evidence about the method on wide schemas. No current-pipeline Step 1 run exists for this dataset (see Section 7), so there is no like-for-like global-condensation comparison point here either.

6.4 Cross-Dataset Synthesis

Five observations hold across the datasets examined.

First, attribute selection is worth doing. On the one dataset with a no-selection baseline, both mechanisms beat matching on all attributes by a wide margin (0.7849 F1 versus 0.9231 and 0.9537), and both do so while spending fewer tokens. Whatever the relative merits of the two mechanisms, the premise underlying both is supported.

Second, global supervised condensation (Step 1) is competitive with or better than the best Step-1.5 sampler on both datasets where both exist, with a far smaller attribute footprint (2 attributes versus 4–5) and 12–18% fewer matching tokens. On Fodors–Zagat it leads on F1 (0.9537 versus 0.9231); on DBLP–ACM the best Step-1.5 sampler edges ahead (0.9605 versus 0.9515) but at higher token cost. For these two datasets a single cheaply-trained global subset captures most of what pair-adaptive selection is trying to achieve — but see Section 5.4, which shows this comparison is not like-for-like.

Third, the only sampler effect that survives repeated seeds is a negative one. Importance-based sampling is reliably the worst sampler on Fodors–Zagat (mean 0.8356 versus 0.8843–0.9222, significant against all four alternatives), and since its profile selection is deterministic this is not one bad draw. Its apparent win on DBLP–ACM (0.9605 versus 0.9066–0.9235) rests on single-seed runs whose reseeding spread is wider than the gap being claimed (Section 5.1), so it cannot currently be set against the Fodors–Zagat result. No sampler can be recommended as a default, and the cross-dataset reversal that earlier drafts reported is not established.

Fourth, the repeated-seed evidence changes what the sampler comparison is a comparison *of*. Among the four non-losing samplers on Fodors–Zagat the mean differences are not statistically distinguishable, but the variances differ by an order of magnitude, and they split precisely along whether the sampler draws its profile pairs or derives them. This suggests the property worth designing for is stability rather than mean accuracy, and it is a property the current similarity-stratified default does not have. The DBLP–ACM **random** sweep is consistent with this: SD 0.0155 over five seeds, against replicate noise of 0.0023.

Fifth, part of what the sampler comparison has been measuring is not attribute selection. Because the selected subset is passed to the matcher in predicted-importance order, reseeding permutes the prompt as well as changing the subset, and on DBLP–ACM permutation alone flips 5.44% of answers against 0.10% for an identical prompt (Section 5.2). Until field order is fixed to a canonical ordering, no Step-1.5 comparison in this report cleanly isolates the variable it intends to test. This is the single highest-value correction available, and it is a small one: sorting the selected attributes into table-column order before condensation would remove the confound without touching the selection logic.

7 Limitations and Missing Data

1. No current-pipeline Step 1 runs for manual, heuristic, or LLM-guided strategies.

Every current-pipeline Step 1 run available uses the supervised strategy; runs for the other three strategies either do not exist, or exist only from a superseded pipeline configuration (different blocking hyperparameters, untracked LLM model). The methodology in Section 3.1

for those three strategies is accurate to the code, but no current-pipeline accuracy/token numbers can be reported for them here. This gap is more consequential than it first appears: *manual* and *heuristic* are the only label-free Step-1 strategies, and therefore the only fair comparison points for Step 1.5 (Section 5.4). Without them, every Step-1-versus-Step-1.5 result in this report pits a label-consuming method against a label-free one.

2. **The Step-1 labeled sample is not held out from evaluation.** The 50 positive pairs drawn for supervised and LLM-guided selection remain in the set the run is scored against, so those strategies are partly evaluated on pairs used to fit them — 45.5% of the ground truth on Fodors–Zagat, 2.2% on DBLP–ACM. What is fitted is a global attribute ranking rather than a per-pair decision, so the effect is indirect, but the Step-1 figures should be read as mildly optimistic.
3. **Amazon–Walmart has no current-pipeline Step 1 run at all,** and only one of the five Step-1.5 samplers (`similarity_stratified`) has been run on the current embedding model and LLM. A full five-sampler sweep and a Step 1 baseline for this dataset would substantially strengthen the cross-dataset comparison in Section 5.
4. **Only Fodors–Zagat has a no-selection baseline** (`--mode full`). DBLP–ACM and Amazon–Walmart have never been run without attribute selection, so the absolute benefit of selection established in Table 9 rests on a single dataset.
5. **The DBLP–ACM reseeding sweep is only one-fifth complete.** Of the thirteen runs attempted, five succeeded (`random`, seeds 42–46) and eight failed on API exhaustion (Section 5.1). The four remaining samplers have no repeated-seed evidence, so the ranking in Table 3 – including the headline result that importance-based sampling wins on this dataset – remains provisional, and is now known to be smaller than the reseeding spread of the one sampler that was measured. Amazon–Walmart has no repeated-seed runs at all.
6. **Failed LLM calls are silently scored as non-matches.** The matcher records an `api_errors` count but `code/main.py` neither aborts nor marks the run as invalid, so a run whose every call failed still writes a complete `run_summary.json` reporting precision, recall and F1. The eight aborted DBLP–ACM runs, and a later Fodors–Zagat Step-1 run, were identified only by inspecting `api_errors` after the fact. The sweep driver now discards such runs and halts, but a direct pipeline invocation still does not, so any manually launched run must be checked before its numbers are used.
7. **Prompt field order is uncontrolled across all Step-1.5 results.** Selected attributes reached the matcher in predicted-importance order, which varies with the profile sample, so every Step-1.5 run varied field order alongside field choice. On DBLP–ACM the order effect is the larger of the two (Section 5.2). It has not been quantified on Fodors–Zagat or Amazon–Walmart, and every Step-1.5 number in this report is affected. The pipeline now imposes a canonical order, but no reported run was produced under it.
8. **Six runs per sampler is a small sample for a variance claim.** The standard deviations in Table 6 are estimated from six observations, and three of the five samplers are deterministic in their sampling, so their spread reflects only matcher-LLM non-determinism and is not a sampling variance at all. The conclusion that stochastic samplers are less stable is supported by an order-of-magnitude gap rather than by a precise estimate of either quantity.

9. **The Step-1.5 selection policy is not scale-free and saturates on wide schemas.** On Amazon–Walmart the cumulative-importance threshold of 0.8 is unreachable within the 5-attribute cap: using the LLM’s own profiled importance vectors, the top 5 of 13 attributes carry on average only 0.612 of the importance mass (maximum 0.789), so reaching 0.8 would require 7.7 attributes on average. The cap therefore binds for 100% of pairs regardless of what the transfer stage predicts, which explains the constant selection size of exactly 5.0 observed in Section 5. This is a property of normalizing importance over a 13-attribute schema, not evidence about the selector’s quality, and it means the Amazon–Walmart run does not currently test adaptive selection in any meaningful sense.
10. **Token savings do not follow attribute savings.** On Fodors–Zagat, Step 1.5 drops 22% of attributes per pair but only 3.8% of tokens, because the selection policy ranks by importance without regard to field length and the dropped fields are the shortest ones. Any efficiency claim for Step 1.5 needs to be stated in tokens rather than attribute counts.
11. **Run-to-run non-determinism.** Two runs of DBLP–ACM’s diversity sampler under an identical configuration (the runs timestamped 20260612.153526 and 20260615.123251, the latter being the one used in Table 3) produced F1 of 0.9087 and 0.9162 respectively. This drift is now partly explained: the two runs predate the fixed blocking seed, so they differ in prompt field order as well as in profile sample. With configuration and seed genuinely held fixed the matcher is close to deterministic (0.10–0.22% of answers differ), so residual single-run drift should be read as an upstream sampling artifact rather than provider-side variance.
12. **No automated leakage check exists.** The Fodors–Zagat `class` attribute was found by manual inspection during analysis, after it had already influenced a full generation of results. DBLP–ACM and Amazon–Walmart have since been checked by hand and are clean, but the pipeline has no guard that would catch a label-valued attribute in a future dataset.

8 Summary

Attribute selection clearly earns its place in the pipeline: on the one dataset where a no-selection baseline exists, both mechanisms improve F1 by 0.13–0.17 while reducing token spend. The comparison between the two mechanisms is less favourable to the thesis’s central contribution than the first generation of runs suggested. Step 1.5’s pair-adaptive selection is the more sophisticated design, but a cheaply-trained Step 1 global condensation using only 2 attributes beats it on Fodors–Zagat and trails it only narrowly on DBLP–ACM, in both cases at substantially lower token cost.

That comparison, however, is not between peers. Step 1’s supervised strategy is fitted on 50 gold matches — 45.5% of the entire Fodors–Zagat answer key — while Step 1.5 consumes no labels at all (Section 5.4). A method given half the ground truth is expected to win; the informative quantity is how small the remaining gap is without any supervision. Settling this requires the two label-free Step-1 strategies, which have never been run on the current pipeline.

The sampler comparison, which was the strongest apparent result in the earlier runs, has shrunk under each round of scrutiny. Removing a leaking attribute from Fodors–Zagat reversed the ranking; sweeping six runs per sampler then showed that most of the remaining ordering was not real, since among the four non-losing samplers the mean differences are not statistically distinguishable; and reseeding DBLP–ACM has now shown that its ranking, too, rests on differences smaller than the run-to-run spread. What survives is one negative result — importance-based sampling is reliably worst on Fodors–Zagat — and one positive one: the samplers separate sharply by variance rather

than by mean, with deterministic samplers varying by 0.002–0.006 F1 across runs against 0.021–0.043 for the two that resample. Stability, not mean accuracy, is the dimension on which the samplers demonstrably differ.

The reseeding work also surfaced two defects in the measurement apparatus rather than in the method. Selected attributes were handed to the matcher in predicted-importance order, so every Step-1.5 run varied the prompt’s field order alongside its field content; on DBLP–ACM, permuting the order while holding the attribute set fixed changes 5.44% of the matcher’s answers, against 0.10% for an identical prompt. And the LLM client declared no token cap, so every request reserved 65,535 tokens of budget against an actual consumption of one, aborting runs while credit remained. Both have been fixed; neither result was available to the runs reported here.

9 Current Status and Remaining Work

The findings above are stable and can be relied on: the Fodors–Zagat label leak and its effect, the variance-versus-mean structure of the sampler comparison, the prompt-order confound and its magnitude, the selection-policy saturation on wide schemas, and the no-selection baseline showing that attribute selection is worth doing. What is *not* settled is the sampler ranking on either dataset.

Two code corrections have been made since the runs reported here and are covered by regression tests, but no experiment has yet been run under them:

1. **Canonical prompt field order** in both selection stages (Section 5.2). This invalidates cross-generation comparison: no pre-fix run may be compared against a post-fix run, so the Step-1.5 tables need regenerating in full rather than extending.
2. **Explicit token caps** on all LLM calls (Section 5.3), removing the spurious budget failures that destroyed nine runs.

Remaining experiments, in priority order, with costs measured from logged token usage at the empirically observed rate of \$0.31 per million tokens:

Work item	Tokens	Est. cost	Runtime
Step-1 <i>manual</i> and <i>heuristic</i> , both datasets	9 M	\$2.79	1.1 h
Fodors–Zagat Step-1 baseline (re-run)	0.7 M	\$0.22	4 min
Fodors–Zagat sampler sweep, 5 samplers $\times$ 5 seeds	20 M	\$6.19	1.7 h
DBLP–ACM <i>full</i> and Step-1 baselines	7.5 M	\$2.32	55 min
DBLP–ACM sampler sweep, 5 samplers $\times$ 5 seeds	98 M	\$30.35	8.2 h
Total to complete the comparison	135 M	\$41.88	12 h
Amazon–Walmart, single run (blocked)	60 M	\$18.58	4.3 h

Amazon–Walmart is listed separately because it is blocked on the selection policy, not on budget: until the cumulative-importance threshold is made scale-free, a run on its thirteen-attribute schema saturates at `max_k` for 100% of pairs and tests nothing (Section 7). Spending on it before that fix would produce another uninterpretable result.

Work is currently halted for a non-technical reason: the OpenRouter account has been exhausted. The balance stands at \$95.00 provisioned against \$95.25 consumed, leaving it \$0.25 overdrawn, and all API calls now return HTTP 402. Completing the outstanding comparison requires approximately \$42 of additional credit; including the Amazon–Walmart run once its policy defect is fixed brings the total to roughly \$61.